INCH-POUND

MIL-R-48427C (AR)
09 November 1992
SUPERSEDING
MIL-R-48427B (AR)
16 May 1990

MILITARY SPECIFICATION

REFLECTOR AND FEED ASSEMBLY: 10556313

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements examinations and tests for the Reflector and Feed Assembly (Drawing 10556313), which is a subassembly of the Antenna Assembly AS-2049/VPS-2, a part of the Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U*S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8588780 - Telescope, Elbow: M109
10556313 - Reflector and Feed Assembly

SPECIAL PACKAGING INSTRUCTION

PS 10556313 - Packaging of Reflector and Feed Assembly

(Copies of other Government documents,- drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity *or* as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10556313, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the reflector and feed assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The reflector and feed assembly shall not suffer damage and shall meet the requirements of 3.4 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibrations as defined by MIL-STD-810 narrow-band random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The reflector and feed assembly shall not suffer damage and shall meet the requirements of 3.4 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The reflector and feed assembly shall meet the requirements of 3.4 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3 .3.4 Temperature.

3.3.4.1 operating. The reflector and feed assembly shall meet the requirements of 3.4 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage) . The reflector and feed assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60"F for a period of 24 hours and +160"F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 when returned to room ambient.

3.3.5 Humidity (non-operating) The reflector and feed assembly shall meet the requirements-of 3.4 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Performance. The reflector and feed assembly shall meet the radio frequency characteristics when tested on an antenna test range which shall provide for field vision ($2D^2/\sim$) . This requires a minimum of 75 feet between transmitting and receiving antennas for the measurements. The reflector and feed assembly shall be tested between the frequency range of 9.15 and 9.25 GHz.

3.4.1 Antenna pattern (without radome. The energy of any side lobe shall be at least 22 decibels (dB) below the main lobe energy.

3.4.2 Antenna pattern (with radome). The energy of any side lobe shall be at least 22 dB below the main lobe energy for the E-plane and not less than 21 dB for the H-plane.

3.4.3 Beamwidth. The beamwidth of the main lobe shall be 3.4 degrees to 4.0 degrees for energy received.

3.4.4 Gain. The reflector and feed assembly shall have a gain of at least 31 dB over an isotropic radiator.

3.4.5 Boresight. The optical axis of the M109 telescope mounted in the boresight clamp and the electrical center of the antenna beam shall be within +1/8 degree.

3.4.6 VSWR. The VSWR of the reflector and feed assembly, when pointed in a non-reflecting direction, shall not be greater than 1.4:1.

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3.5 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General Provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4) .

c. Special sampling inspection (see 4.5) .

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90"F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies components and test specimens listed below in the quantities indicated.

Name	Drawing	quantity
Reflector and Feed Assembly	10556313	3

4.3.2 Inspections to be performd. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements~ the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials

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and methods, in accordance with the same drawings, same drawing revisions same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP) . In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool . control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum the level Of qualitY assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4-.4.2.3 and 4.5 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests *in* Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCeDURe
Critical	None	N/A
Major 101. Electrical	3.4	4.7.2
Minor	None	[N/A

4.5 Special sampling. One reflector and feed assembly out of the first 50 produced and one out of each 100 produced thereafter shall-be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table 11. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical	None	N/A
Major 102, Vibration 103. Shock 104. Altitude 105. Temperature 106. Humidity (non-operating)	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Goverment designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

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4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table III (Test equipment list) shall be utilized in the performance of the applicable test methods of inspection as specified in 4.7.

TABLE 111. Test equipment list.

1- NUMBER	DeScriPTION
1. Signal generator	Frequency: 9.15 to 9.25 GHz. RF power output: 1 and 100 xnW. Modulator: AM modulation to 50% at 1 KHz , tunable ± 10%.
2. SWR meter and detector	Sensitivity: 0.15 pV, tuned to 1 KHz; calibrated to read VSWR with a square-law detector.
3. Slotted line and detector	Frequency: 9.15 to 9.25 GHz. Residual SWR: Less than 1.05:1. Sensitivity: Sufficient to provide greater than 1 UV at 1 KHz. Square law over greater than 20 dB.
4. Attenuator	Attenuation: lo dB ± 5% SWR : 1 to 1.1. Frequency: 9.2 GHz.
5. Variable attenuator	Attenuation: O to 22 dB ± 5%. Frequency: 9.2 GHz.
6. Directional coupler	Coupling factor: 20 ± 1 dB. SWR: 1 to 1.1.
7. Frequency meter	Frequency: 9.15 to 9.25 GHz. Accuracy: * 1%
8. Standard gain horn	Gain between 15 and 25 dB at 9.2 GHz.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
9. Antenna test range	Distance between antennas: 75 ft. Std. antenna gain: Sufficient to record VPS-2 antenna pattern with minimum 30 dB dynamic range. Pedestal repeatability: ± 1 min. Pedestal azimuth travel: 360 degrees Pedestal elevation travel: * 20 degrees from horizon. Frequency: 9.15, 9.2 and 9.5 GHz Recorder: Signal vs. direction * 1 min. Visual boresight: Target displaced in same direction from center of transmitting antenna by equal amount that boresight clamp is displaced from center of antenna assembly (11.25 ± .01 inch).
10. Test fixture (oPtional)	Fixture to mount antenna and return antenna in both planes, to within ± minute of center between two electrical measurements.
11. Telescope elbow	M109, 8588780

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

-- 4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I Of MIL S~810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph 11-2.1 apply. The reflector and feed - assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attach ment . After vibration, the reflector and feed assembly shall be tested as specified in 4.7.2 herein to determine compliance with 3.3.1.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± lo%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the reflector and feed assembly shall be tested as specified in 4.7.2 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating) This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operatina) . This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 for compliance with 3.3.4.

4.7.1.4.2 Low (operatina and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure"IIS, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard arribient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

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4.7.2 Performance. The tests for antenna pattern, beamwidth, gain, boresight, and VSWR (4.7.2.1, 4.7.2.2, 4.7.2.3, 4.7.2.4, and 4.7.2.5) shall be performed on all reflector and feed assemblies.

4.7.2.1 Antenna pattem. The antenna pattern is to be obtained as follows:

a. Connect signal generator (Table III, item 1) to antenna test range transmitting antenna (Table III part of Item 9) through variable attenuator (Table III, item 5) and adjust frequency for 9.2 GHz.

b. Use test fixture (Table III, item 10) to mount reflector and feed assembly to pedestal and receiver of antenna test range (Table IIIt part of Item 9) and Point reflector and feed assembly toward test transmitting antenna. Adjust antenna test range equipment controls for apeak on test range recorder while turning, in all directions, reflector and feed assembly under test for maximum recorder deflection.

c. Adjust test range equipment for full recorder deflection.

d. Rotate reflector and feed assembly under test to 180 degree (in azimuth) from transmitting antenna and make a 360 degree azimuth recording of reflector and feed assembly under test. Return reflector and feed assembly to peak deflection position of antenna test range and increase variable attenuator setting by 22 dB.

Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the reflector and feed assembly to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the reflector and feed assembly as recorded above shall not be greater than the 22 dB down reference level established in step d.

f. Repeat step a to obtain full deflection. Rotate reflector and feed assembly under test in a vertical direction to 20 degrees from horizontal. Record pattern in elevation $\pm$ 20 degrees from horizon. Return reflector and feed assembly to peak deflection of recorder and increase variable attenuator attenuation by 22 dB. -

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9. Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the reflector and feed assembly vertically to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the reflector and feed assembly as recorded above shall not be greater than the 22 dB down reference level established in step f.

h. Repeat steps a through g for frequencies 9.15 and 9.25 GHz.

4.7.2.2 Beamwidth. The beamwidths will be obtained by repeating 4.7.2.la for 9.2 GHz.

a. Operate recorder to obtain expanded directional pattern (i.e., 6 degree to 60 degree) .

b. Start azimuth pattern at least 3 degree deflection from peak obtained above and record at lease 3 degree past peak.

c. Return reflector and feed assembly to peak and adjust variable attenuator to obtain 3 dB attenuation.

d. Use this level to mark antenna pattern with 3 dB reference.

e. The 3 dB beamwidth as obtained shall be between 3.4 degree and 4.0 degree.

f. Repeat steps a through e except rotation will be in elevation.

9* The 3 dB beamwidth as obtained shall be between 3.4 degree and 4.0 degree.

h. Repeat steps a through h for frequencies of 9.15 and 9.25 GHz.

The 3 dB beamwidth as obtained shall be between 3.4 degree and 4.0 degree.

4.7.2.3 Gain. The gain shall be obtained by performing 4.7.2.la for 9.2 GHz using antenna test range or optical test fixture (no recording required).

Adjust the antenna test range equipment to obtain a variable attenuation of at least 3 dB greater than 31 dB, minus the “standard gain horn” gain when full recorder deflection or similar indication is obtained.

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b. The reflector and feed assembly under test is replaced with standard gain antenna and variable attenuator adjusted to obtain full recorder deflection.

c. The change in variable attenuator setting plus the gain of the standard gain antenna shall be no less than 31 dB.

4.7.2.4 Boresight. The boresight of the reflector and feed assembly shall be accomplished using the antenna test range (Table III, item 9) or holding fixture which allows movement of the reflector and feed assembly, ± degree from center and return to center within ± 1 minute from electrical measurement.

The reflector and feed assembly shall be adjusted for electr?;al center peak by repeating 4.7.2.la for 9.2 GHz using antenna test range or test fixture.

b. Adjust variable attenuator to increase attenuation by 3 dB and record level of recorder deflection.

c. Return variable attenuator to value before 3 dB increase.

d. Rotate reflector and feed assembly until recorded level of 3 dB above is passed in azimuth.

Return reflector and feed assembly back to obtain 3 dB reference above (above operations to overcome backlash).

f. Note reading or indication and continue rotation through peak and to next 3 dB reference. Note reading.

g. Move reflector and feed assembly to halfway between the two 3 dB readings taking care to go past desired reading and return so approach is from same direction as before.

h. Repeat step d for elevation. Place M109 telescope (Table III, item 11) in boresight telescope holder and observe boresight target in cross hairs.

4.7.2.5 VSWR. Standing wave measurements are obtained by connecting signal generator (Table III, item 1) output to slotted line and detector (Table III, item 3) with attenuator (Table III, item 4) between item 1 and 3.

a. Connect SWR meter (Table III, item 2) to detector output .

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b. Connect output of slotted line to reflector and feed assembly under test and aim antenna in a nonreflecting direction.

c. Turn equipment on and allow to stabilize.

d. Adjust frequency of signal generator to 9.2 GHz.

e. Adjust the modulation frequency of signal generator to obtain a peak reading of the SWR meter.

f. Adjust the signal generator attenuator level and slotted line probe for a full scale reading on the SWR meter and so that all measurements are made within the square law region of the slotted line detector.

g. Obtain the SWR as indicated on the SWR meter by movement of the slotted line probe for maximum to minimum voltage levels.

h. Repeat steps 3 through g for 9.15 and 9.25 GHz. SWR shall be less than 1.4 to 1.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 10556313. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reflector and feed assembly is used for the transmission and reception of pulsed X-band radar signals. The reflector and feed assembly is incorporated into the Antenna System, AS-20491VPS-2, a part of the Radar Set ANIVPS-2.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.2).

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c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g* Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection requiRement designs for approva 1. Submit copies of designs as required to: Commander, U.S. Army C,ATTN: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island
a Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Sub-ject term (kev word) listing.

Antenna Assembly
BeamWidth
Boresight
Gain
Radar Set AN/VPS-2
VSWR

6.6 Changes from Previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A872)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-48427	09 NOVEMBER 1992

3. DOCUMENT TITLE: REFLECTOR AND FEED ASSEMBLY: 10556313

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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